Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/11/2005
PI Name:	Meystre, Pierre Ph.D.		
Project Title:	Atom optics in controlled and microgravity environments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/30/2001	**End Date:**	04/30/2005
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth"s gravity		

Task Description:	The goal of this project is to carry out theoretical investigations of atom optics in environments characterized by controlled and adjustable accelerations and in field-free situations. The main thrusts of the research are: the theoretical study of nonlinear atom optics, and in particular matter-wave mixing phenomena, in accelerated frames; the investigation of the evolution of matter-wave coherence in accelerated atomic de Broglie waves, with applications in rotational and gravitational sensors; the detailed theoretical analysis of atom holography in microgravity environments, including the use of accelerated beams for the reconstruction process; and a broad theoretical investigation of field-sensitive situations that cannot realistically be performed in earhtbound laboratories. Atom optics is on the verge of becoming an engineering science with broad potential applications, in particular in rotation sensors, gravimeters and gravity gradiometers. Other possible applications include nanofabrication, atom holography, etc. In many cases, the atomic source will be a quantum degenerate gas, such as a Bose condensate --- the atom optical analog of a laser --- or, in the future, a Fermi-degenerate atomic system. Practical devices require however matter waves of short wavelengths, which can be realized by particle acceleration. In addition, external fields used e.g. to compensate earth gravity can lead to random frequency shifts and decoherence that must be avoided in precision measurements. For these and other reasons, it is important to study atom optics in controlled and microgravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In hte last year or so, we have made several major advances on the use of atom optics in microgravity environments. In view of the new recent thrust of NASA on Moon-March exploration, we have also reoriented our work to concentrate more directly on navigation and guidance using matter-wave interferometers and devices. One important development is the study of the use of fermionic matter-wave fields in atom interferometry. It had long been believed that Bose-Einstein condensates would play for atom optics a role similar to that of lasers in optics. However, the situation is not so simple, because in contrast to light, matter-wave fields are self-interacting. Cold collisions result in nonlinear phase shifts that can severely limit the accuracy of BEC-based atom interferometers, gravity gradiometers, etc. In contrast, at ultracold temperatures, quantum-degenerate fermionic atoms do not interact, a direct consequence of the Pauli Exclusion Principle. On the other hand, this same Pauli Exclusion Principle prohibits one from realizing monoenergetic beams of quantum-degenerate fermionic atoms, so the question is whether this limitation is more restrictive than the collisional phase shifts plaguing bosonic interferometers and gyros. We have initiated a study of this question, and our results so far indicate that in many circumstances, fermions can actually be significantly more advantageous. This surprising result is somewhat related to the situation in optics, where there are a number of circumstances where white-light interferometry can be vastly superior to the use of monochromatic sources. Encouraged by these results, we are now moving full speed ahead in fermionic atom optics, a difficult task theoreticall, but with high potential pay-off. In other work, but not ready for prime-time yet, we are also looking into the use of hybrid optical-matter wave optics for gravitational sensors and gravity-gradiometers. In another recent development, we have started to investigate the potential of ultracold heteronuclear diatomic molecules for matter-wave interferometry. The fact that they are polar makes it possible to trap them in centimeter-size storage rings for tens of round trips and as a result, offers the potential for orders-of-magnitude improvements over neutral atoms for inertial sensors.
Bibliography Type:	Description: (Last Updated: 02/24/2006)
Articles in Peer-reviewed Journals	C. P. Search, C. P.; Zhang, W.; Meystre, P. "Inhibiting three-body recombination in atomic Bose-Einstein condensate" Phys. Rev. Lett. 92, 140401 , Apr-2004
Articles in Peer-reviewed Journals	Search, C. P.; Miyakawa, T.; Meystre, P. "A lattice array of molecular micromasers" Optics Express 12, 30 , Jan-2004
Articles in Peer-reviewed Journals	Search, C. P.; Meystre, P. "Molecular matter-wave amplifier" Phys. Rev. Lett. 93, 140405 (2004) , Oct-2004
Articles in Peer-reviewed Journals	Miyakawa, T; Search, C. P.; Meystre, P. "Phase coherence in a driven double-well system" Phys. Rev. A 70, 053622 (2004) , Nov-2004
Articles in Peer-reviewed Journals	Jaaskelainen, M; Zhang, W; Meystre, P. "Limits to phase resolution in matter-wave interferometry" Phys. Rev. A 70, 063612 , Dec-2004
Awards	Meystre, P. "Elected Fellow of the American Association for the Advancement of Science" Oct-2004
Presentation	Meystre, P. "Nonlinear matter-wave optics of atoms and molecules" Invited talk, Binational Consortium of Optics Winter School, Tucson, AZ Jan-2004
Presentation	Meystre, P. "Fermionic atom optics" Invited talk, 2004 Cold Gases Program, Kavli Institute for Theoretical Physics, Santa Barbara, CA Jun-2004
Presentation	Meystre, P. "Micromaser-like generation of molecular fields" Invited talk, RIAO/OPTILAS 2004, Isla de Margarita, Venezuela Nov-2004
Presentation	Meystre, P. "Nonlinear optics of atoms and molecules" nvited talk, Quantum Optics II, Cozumel, Mexico Dec-2004
Presentation	Jaaskelainen, M; Zhang, W.; Meystre, P. "Phase resolution in matter-wave interferometry" DAMOP-DAMP Annual Meeting, Tucson, AZ May-2004
Presentation	Miyakawa, T.; Search, C.; Meystre, P. "Phase coherence in molecular micromasers with interwell tunnelling" DAMOP-DAMP Annual Meeting, Tucson, AZ May-2004

Presentation	Meiser, D.; Search, C.; Meystre, P. "Diffraction of ultracold fermions off quantized fields: standing waves versus running waves" DAMOP-DAMP Annual meeting, Tucson, AZ May-2004
Presentation	Search, C.; Zhang, W.; Meystre, P. "Inhibiting three-body recombination in atomic Bose-Einstein condensates" DAMOP-DAMP Annual Meeting, Tucson, AZ May-2004